Image tensors and channels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every shape carry an explicit schema.

Use complete pixels,channel planes and known-index checks;separate axis conversion,fixed scaling and colour averaging.

Keep one habit: Assess independent plane values,green index,batch shape and count with checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Read a pixel using declared row column and **channel** axes

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Different **input**s can map to the same representation. The number of components alone does not guarantee that the original values can be recovered.

Opening retrieval

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an **input** through a hidden representation and a decision rule

Reference example, not the only acceptable wording: Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how starting values affect a fixed **training** budget

Reference example, not the only acceptable wording: Teacher explanation: States are -2, 0 and 1. Final **training** MSE is 2.5. Each step halves the signed distance from 2.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Run a controlled comparison of learning rates

Reference example, not the only acceptable wording: Teacher explanation: First v = 0.5 with MSE 5.625. Second gradient -7.5 gives v = 0.875 and MSE 3.1640625. Distance is multiplied by 0.75 each step.

Pictured end goal

By the end, I can explain image **tensor**s and **channel**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Learned representations

Initialisation and reproducibility

Learning rate experiments

For the compact alternative schedule, use two 45-minute sessions. Draw the four-pixel image and label row, column and **channel** before writing nested brackets. Keep RGB ordering, zero-based indices and the 0 to 255 contract explicit.

Distinguish an axis permutation from regrouping, and fixed scaling from fitted pre**process**ing. No external image files or student photographs are needed.

Outcomes

Read a pixel using declared row column and **channel** axes

Convert **channel** layout without changing pixel meaning

Distinguish fixed **value scaling** from information losing transformations

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach axis conversion with complete named-pixel checks

> Keep main2×2,pictured2×1and independent1×2images distinct.

Main contract,complete planes/indices and guided key

A small fictional RGB image has height 2 pixels, width 2 pixels and three **channel**s ordered red, green, blue. Store it as image[row][column][**channel**], abbreviated HWC.

Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are **channel**s 0, 1 and 2. Each stored component is an integer from 0 to 255.

The shape is 2 by 2 by 3. There are four pixels and twelve **channel** values. The blue component of the bottom-left pixel is image[1][0][2] = 255.

A single **channel** is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three **channel**s here; other formats can include additional **channel**s.

Declare the format instead of assuming every three-dimensional array means the same thing.

A **channel**-first layout uses image[**channel**][row][column], abbreviated CHW. Our image then has shape 3 by 2 by 2. A correct conversion preserves the rule chw[c][r][k] = hwc[r][k][c].

The code assumes a nonempty rectangular image with exactly three values per pixel.

Reading nested values in order and merely regrouping them into a new shape does not perform this axis conversion. The first four flattened HWC values are [255, 0, 0, 0].

A naive regrouping makes these the first **channel** plane, but the correct red plane is [[255, 0], [0, 255]]. Shape agreement alone cannot detect this mistake.

CHW channel index / name	Row0 values(column 0 then1)	Row1 values(column 0 then1)
0 /red	[255, 0]	[0, 255]
1 /green	[0, 255]	[0, 255]
2 /blue	[0, 0]	[255, 255]

Same measurement	HWC index	CHW index	Preserved value
Top-right green	[0][1][1]	[1][0][1]	255
Bottom-left blue	[1][0][2]	[2][1][0]	255

Same measurement	HWC index	CHW index	Preserved value
Bottom-right red	[1][1][0]	[0][1][1]	255
Top-left blue	[0][0][2]	[2][0][0]	0

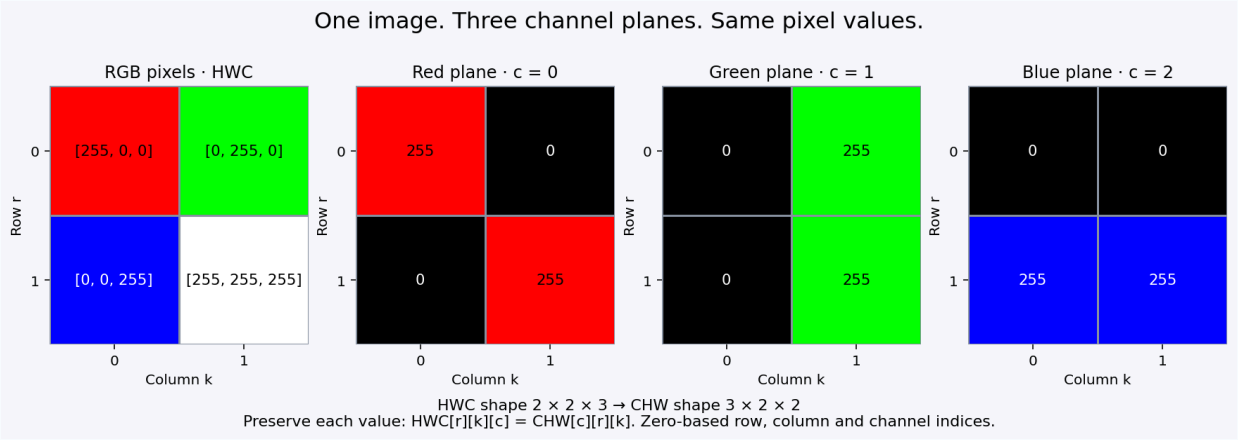
The complete converter assumes a nonempty rectangular image with exactly three valid raw channel values per pixel. It checks a known measurement and round-trip reconstruction for the supplied image, not every possible malformed file. CHW conversion reads the same value with a different axis order. Merely reshaping consecutive flattened values gives a wrong red plane even when dimensions look right.

Keep the RGB channel names, raw 0–255 range and traversal order with the shape.

Guided: top-right is [0, 255, 0] and bottom-right is [255, 255, 255]. The top-right green value is hwc[0][1][1] or chw[1][0][1], both 255. Exit: shape alone misses both errors.

Check the top-left RGB pixel is red before scaling and that a raw value 255 becomes exactly 1 under one declared division, rather than approximately 0.00392.

Labelled image and all three channel planes



Separate pictured values and full batched index

Move colour values between **tensor** layouts without changing the pixels.

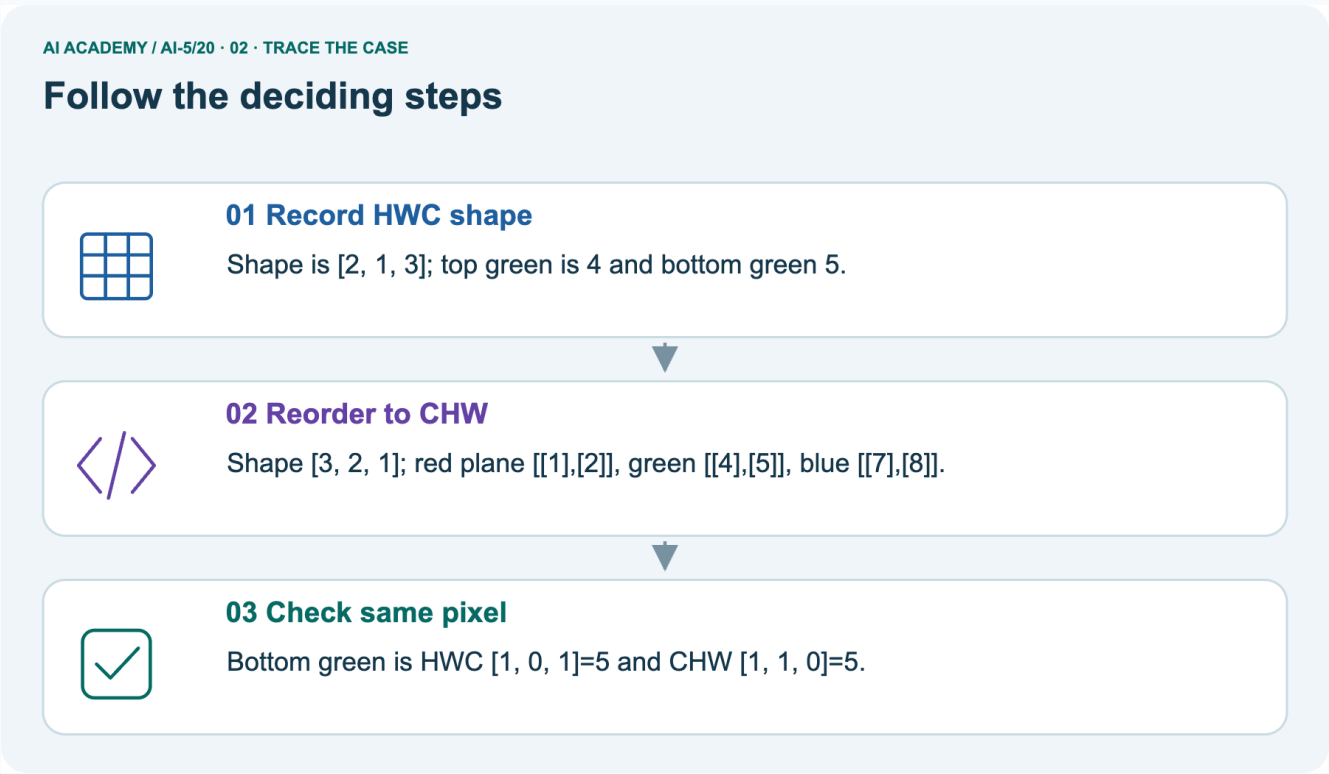
Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate two-row, one-column RGB image:top[1, 4, 7], bottom[2, 5, 8]. HWC shape 2×1×3 becomes CHW3×2×1, with red[[1], [2]], green[[4], [5]], blue[[7], [8]]. Bottom green stays 5at HWC[1][0][1]and CHW[1][1][0]. Three such images in NHWC have shape 3×2×1×3; bottom green in the first image is batch[0][1][0][1].

These small values are raw classroom examples, not the main red/green/blue/white image's255 values.

Trace preserved measurement through the new batch axis



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record HWC shape	Shape is [2, 1, 3]; top green is 4 and bottom green 5.
2. Reorder to CHW	Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].
3. Check same pixel	Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

Change: Add a batch containing three identical images in NHWC layout.

Prediction: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Tensor	A numerical array with a declared set of axes
Channel	One component plane of an image such as red green or blue
Axis order	The sequence of meanings assigned to array dimensions
Value scaling	A declared numerical conversion applied to stored values

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Image tensors and channels

The facts we will use

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/20

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.

2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Teacher check: Teacher explanation: The incorrect plane is [[255, 0], [0, 0]]. The correct red plane is [[255, 0], [0, 255]]. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Channel**s are component planes at matching spatial positions in one image.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

NHWC [0, 1, 0, 1]=5: image 0, row 1, column 0, green **channel** 1. The value is unchanged by batching. Changed-case check: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Changed-case reasoning: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use three labelled paper grids and a coordinate ledger; provide colour names so success does not depend on colour vision. Allow fraction answers for scaling.

Extension: convert a rectangular image HWC to CHW and back, checking every coordinate, then construct two different colours with the same classroom **channel** mean.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve information loss and label the four image positions with zero-based indices.
7 to 18 minutes	Count pixels and components, then use W1 to read a full pixel and one channel.
18 to 32 minutes	Trace the three channel planes and run or hand-trace the optional code.
32 to 45 minutes	Use W2 to expose naive regrouping with the bottom-right red component.
Session 2 start to 8 minutes	Reconstruct the guided pixels and locate the same green value in both layouts.
8 to 20 minutes	Use W3 to compare fixed scaling with colour averaging and discuss double scaling.
20 to 35 minutes	Collect W4 independently, checking the non-square shape and the leading batch axis.
35 to 45 minutes	Use W5 and the exit response to plan exact-value checks, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main complete planes/indices/probes supplied. W2 wrongred[[255, 0], [0, 0]], correct[[255, 0], [0, 255]], mismatchr 1c1. W3 fixedscaling[1, 128/255, 0], simplemeans 85/85 losecolour. W5 knownrawtop-leftmustberedandraw 255 mustscaleonce1, not.003922. IndependentHWC1×2×3; planesR[[10, 40]], G[[20, 50]], B[[30, 60]], CHW3×1×2. GreenatHWC[0][1][1]/CHW[1][0][1]=50. FourcopiesNHWC4×1×2×3, total 24.

State exact axes and counts, not only array lengths.

Diagnose the current misconceptions

Three **channel**s mean three images: **Channel**s are component planes at matching spatial positions in one image.

Reshaping automatically changes axis meanings correctly: Axis permutation must preserve coordinate correspondence; regrouping a flattened list can mix positions and colours.

Every conversion to a smaller number preserves information: Exact fixed scaling differs from rounding and **channel** averaging, which can collapse distinct values.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

Teacher explanation: HWC shape is 1 by 2 by 3. CHW planes are [[10, 40]], [[20, 50]], [[30, 60]], giving shape 3 by 1 by 2. hwc[0][1][1] and chw[1][0][1] both equal 50. Batch shape is 4 by 1 by 2 by 3, containing 24 values.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Read a pixel precisely

For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve **channel** values.

W2 Audit an axis conversion

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Teacher explanation: The incorrect plane is [[255, 0], [0, 0]]. The correct red plane is [[255, 0], [0, 255]]. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

W3 Check scaling and lost colour

Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

Teacher explanation: Scaling gives [1, 128/255, 0]. Both RGB means are 85, so averaging loses the red versus green distinction. Exact fixed scaling preserves the triplet distinction.

W5 Design revealing pipeline checks

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

Teacher explanation: Before scaling, the top-left pixel must be [255, 0, 0], not [0, 0, 255]. A known raw component of 255 must become 1 after scaling once; approximately 0.00392 indicates division twice.

Test named values and positions as well as dimensions.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: top-right is [0, 255, 0] and bottom-right is [255, 255, 255]. The top-right green value is hwc[0][1][1] or chw[1][0][1], both 255. Exit: shape alone misses both errors.

Check the top-left RGB pixel is red before scaling and that a raw value 255 becomes exactly 1 under one declared division, rather than approximately 0.00392.

W1 Read a pixel precisely

W2 Audit an axis conversion

W3 Check scaling and lost colour

W4 Convert a new image independently

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

Teacher explanation: HWC shape is 1 by 2 by 3. CHW planes are [[10, 40]], [[20, 50]], [[30, 60]], giving shape 3 by 1 by 2. hwc[0][1][1] and chw[1][0][1] both equal 50. Batch shape is 4 by 1 by 2 by 3, containing 24 values.

W5 Design revealing pipeline checks

Independent assessment and fresh retry

W4 requires three planes, both coordinate references to value 50, NHWC shape 4 by 1 by 2 by 3 and 24 values. Fresh retry: a two-row one-column image contains [3, 6, 9] above [12, 15, 18]. HWC shape is 2 by 1 by 3.

CHW planes are [[3], [12]], [[6], [15]], [[9], [18]]. The bottom blue component is hwc[1][0][2] = chw[2][1][0] = 18. A batch of three has NHWC shape 3 by 2 by 1 by 3 and 18 values.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled image **tensor**s and **channel**s **evidence** sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and **channel** axes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use three labelled paper grids and a coordinate ledger; provide colour names so success does not depend on colour vision. Allow fraction answers for scaling.

Extension: convert a rectangular image HWC to CHW and back, checking every coordinate, then construct two different colours with the same classroom **channel** mean.

Week 21 introduces convolution as a local operation with explicit neighbourhoods and boundary rules.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/21, ai-5/23, ai-5/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Read a pixel using declared row column and **channel** axes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane.

Identify a position where they disagree.

Reasoned practice answer: Teacher explanation: The incorrect plane is [[255, 0], [0, 0]]. The correct red plane is [[255, 0], [0, 255]]. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

Check every learner before discussion: Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Channel**s are component planes at matching spatial positions in one image.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add a batch containing three identical images in NHWC layout. Require a **prediction** and a reason before comparing. Expected result: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

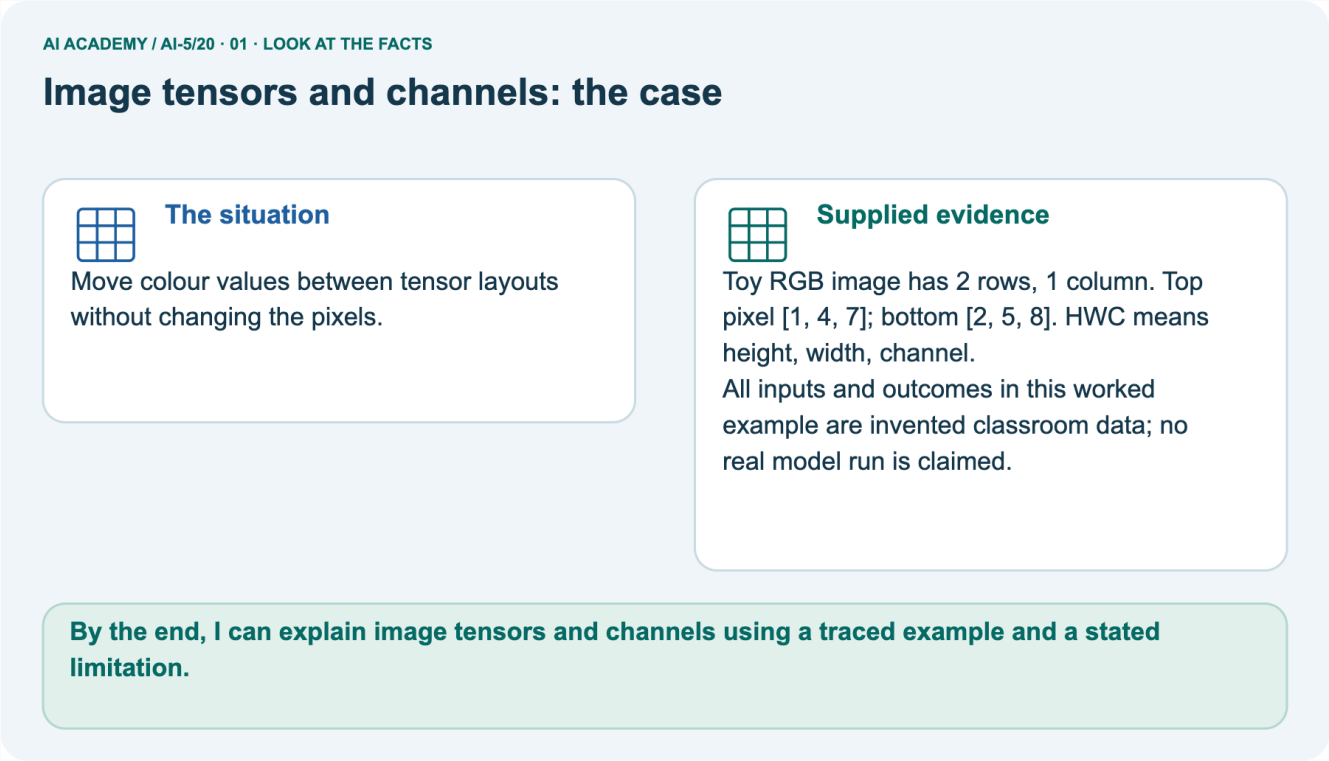
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Image tensors and channels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

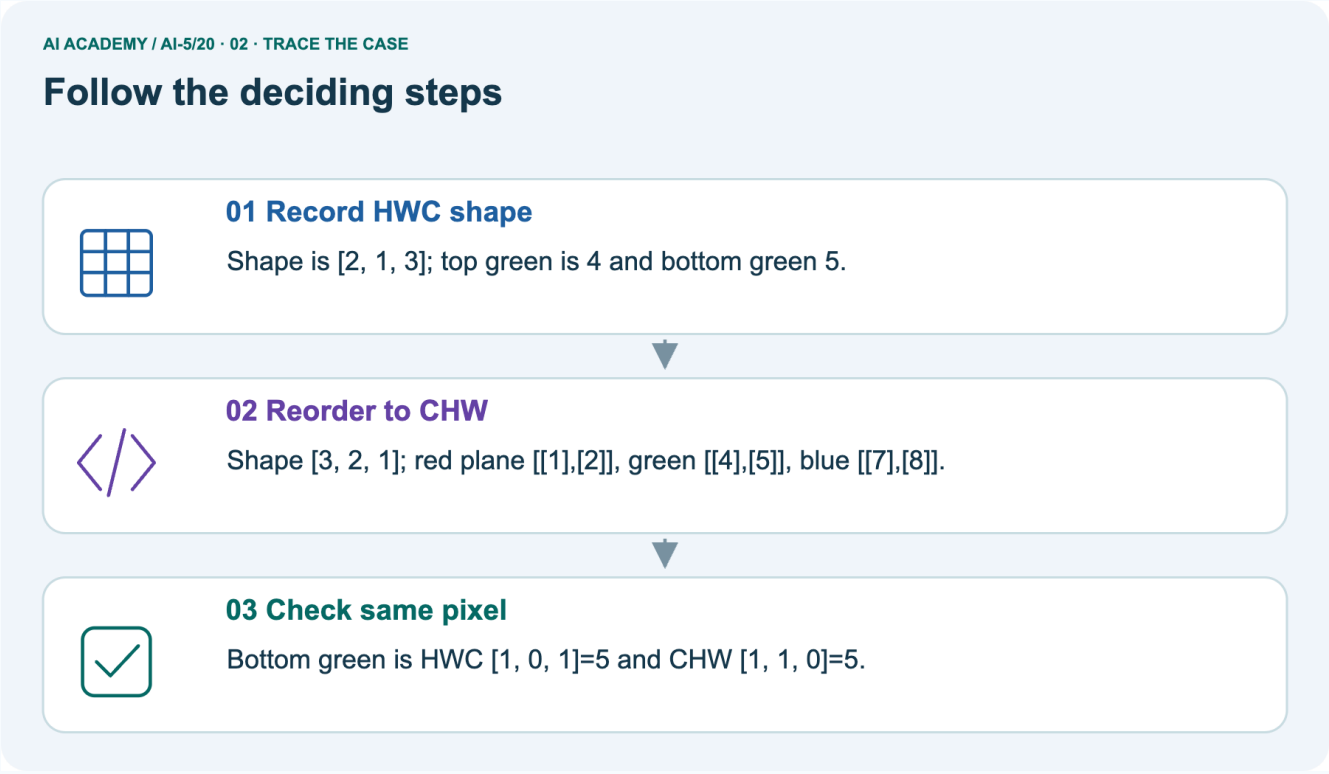
Move colour values between **tensor** layouts without changing the pixels. Supplied facts: Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



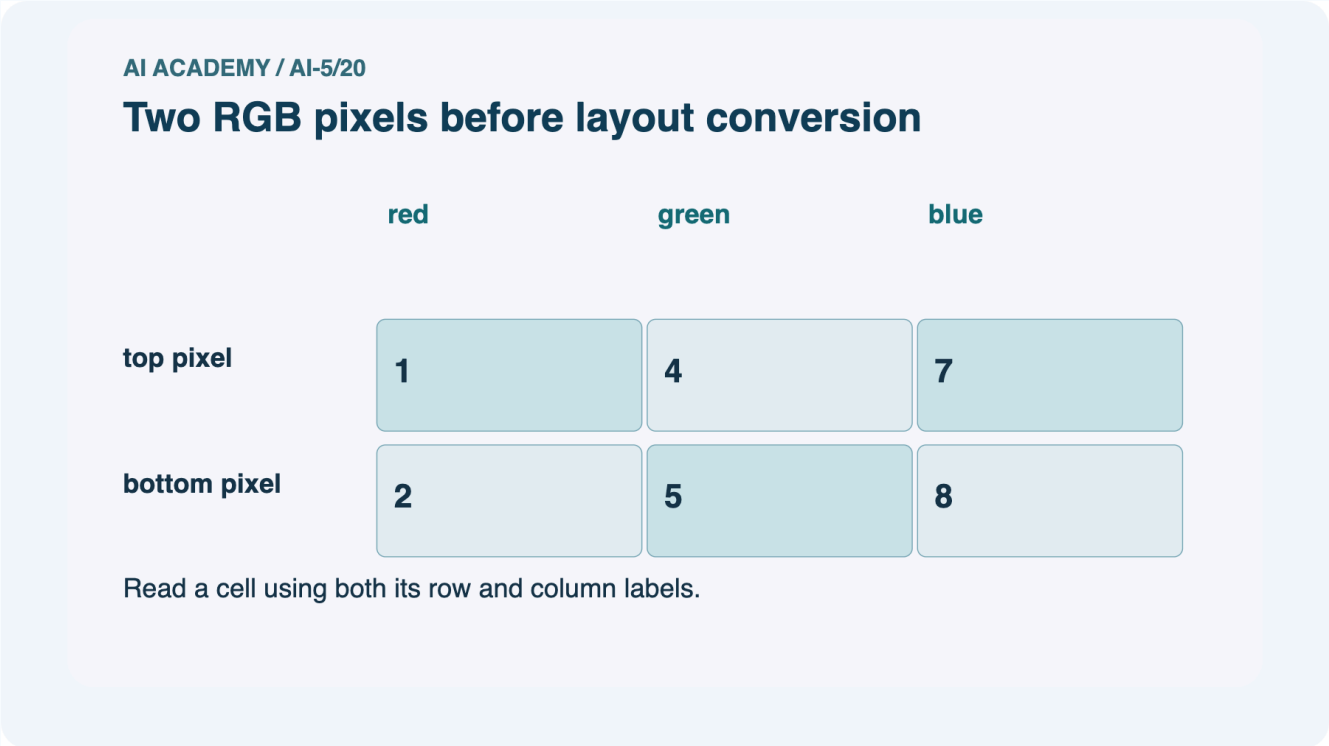
Follow the labelled picture with the matching explanation. Open the vector diagram

Record HWC shape: Shape is [2, 1, 3]; top green is 4 and bottom green 5. → Reorder to CHW: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]]. → Check same pixel: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two RGB pixels before layout conversion



Follow the labelled picture with the matching explanation. Open the vector diagram

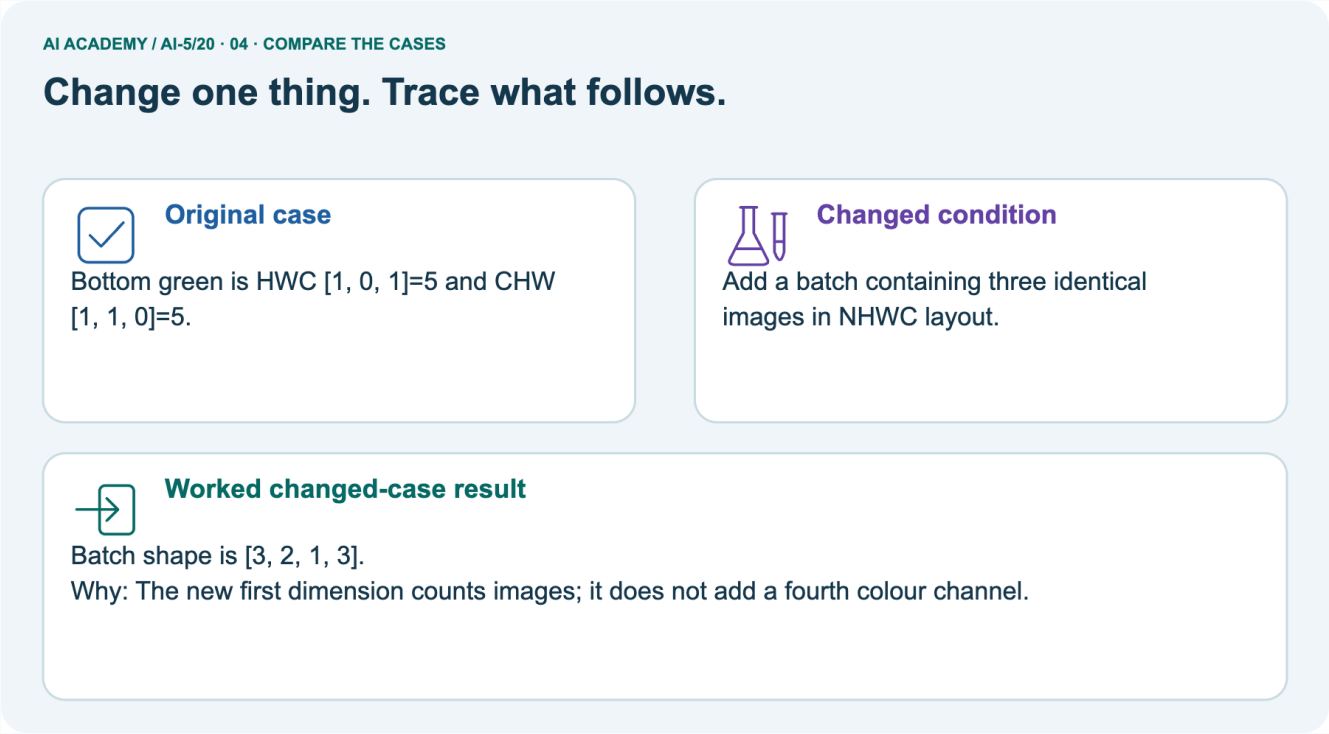
Two RGB pixels before layout conversion. top pixel: red = 1, green = 4, blue = 7. bottom pixel: red = 2, green = 5, blue = 8.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. Changed condition: Add a batch containing three identical images in NHWC layout. Expected result: Batch shape is [3, 2, 1, 3].

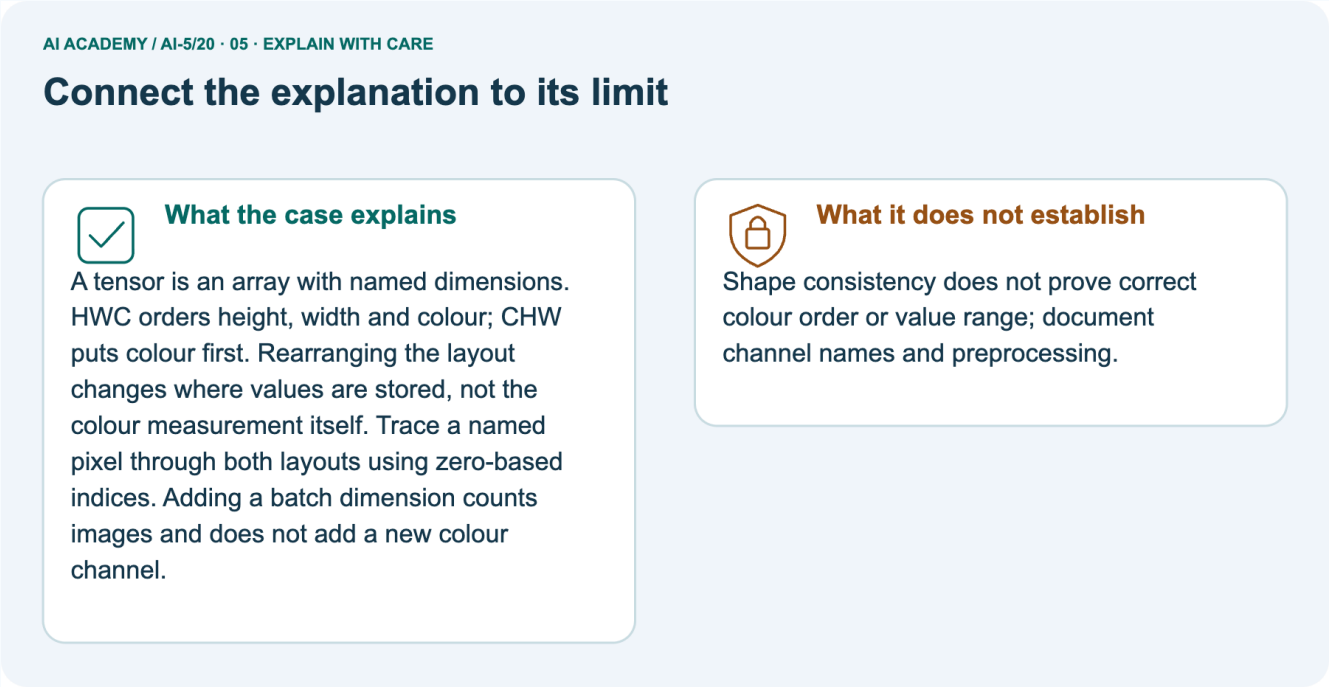
Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The new first dimension counts images; it does not add a fourth colour **channel**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

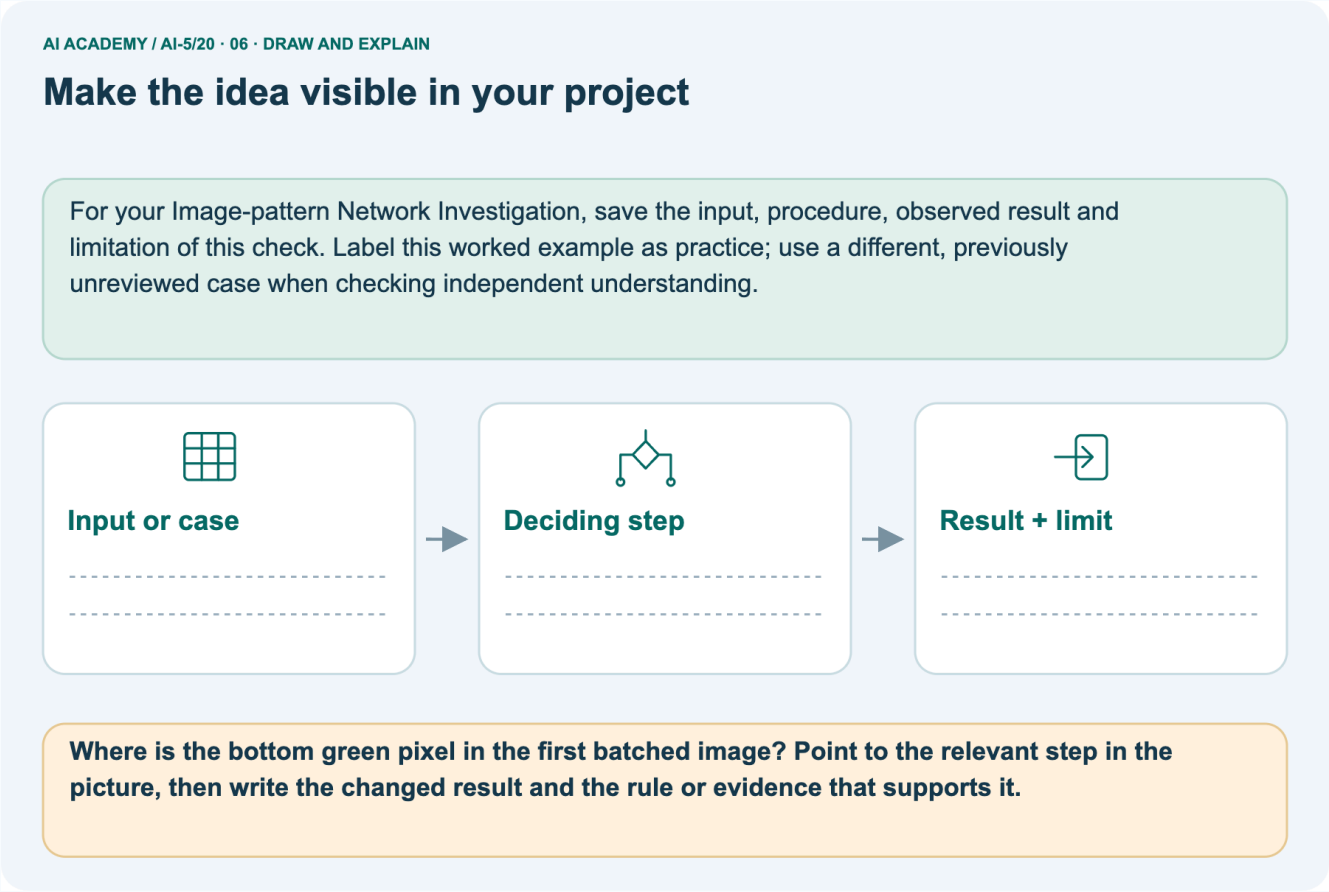
Limit: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: NHWC [0, 1, 0, 1]=5: image 0, row 1, column 0, green **channel** 1. The value is unchanged by batching. Changed-case check: Batch shape is [3, 2, 1, 3].

The new first dimension counts images; it does not add a fourth colour **channel**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Image tensors and channels: the case

02. Follow the deciding steps

03. Two RGB pixels before layout conversion

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.